Binary Format Header

Field	Type	Size (bytes)	Value	Description
Signature	char[4]	4	“LTRF”	LightTools Binary file identifier
Major Version	integer	4	2 or 3	3.0 is used when polarization data is included
Minor Version	integer	4	0	
DataType	integer	4	0 or 1	0 if data is "Radiant Power" in Watts, 1 if data is "Luminous Flux" in Lumens
FarField	integer	4	0 or 1	0 if ray coordinates are written in the file, 1 if starting point coordinates are not written out
ColorInfo	integer	4	0 or 1 or 2	0 - no color info, photopic format 1 - CIE tristimulus values written out 2 - Wavelength info is written out
LengthUnits	integer	4	0 - 7	0 - system units 1 - millimeters 2 - centimeters 3 - meters 4 - inches

				5 - feet 6 - micrometers 7 - nanometers
Origin X	single	4		X origin of the source
Origin Y	single	4		Y origin of the source
Origin Z	single	4		Z origin of the source
Flux/Power	single	4		total power/flux of the source in Watts or Lumens as specified by DataType field
Polarization	integer	4	0 or 1	0 - no polarization data 1 - Export Stokes parameter and direction vector

Binary Format Ray Records

For a ray data file in binary format, the ray records immediately follow the header. They use single precision floating numbers (4 bytes) for ray information. Otherwise, their structure is the same as for the text format. Wavelength, if written out, is always in nanometers.

Example 1: ColorInfor is 2 (wavelength) and FarField is 0 (write out ray coordinates). Record size is 32 bytes.

xPos yPos zPos LDir MDir NDir RelativeFlux Wavelength

Example 2: ColorInfor is 1 (tristimulus) and FarField is 0. Record size is 36 bytes.

xPos yPos zPos LDir MDir NDir CIE_X CIE_Y CIE_Z

Example 3: ColorInfor is 1 (tristimulus), FarField is 1, and Polarization is 1. Record size is 52 bytes.

LDir MDir NDir CIE_X CIE_Y CIE_Z S0 S1 S2 S3 polDirX polDirY polDirZ

Binary Format Footer

The binary format for a ray data file shall always end with the closing string “LTRFEND” of type char, size seven bytes.